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Traffic emergencies occur when two vehicles are about to collide.

Vehicle emergencies occur when tires, brakes, or other critical parts fail.

Following the safety practices in this manual can help prevent emergencies. But if an emergency does happen, your chances of avoiding a crash depend on how well you take action. Actions you can take are discussed below.

2.17.1 – Steering to Avoid a Crash

Stopping is not always the safest thing to do in an emergency. When you do not have enough room to stop, you may have to steer away from what is ahead. Remember, you can almost always turn to miss an obstacle more quickly than you can stop (however, top-heavy vehicles and tractors with multiple trailers may flip over).

Keep Both Hands on the Steering Wheel. In order to turn quickly, you must have a firm grip on the steering wheel with both hands. The best way to have both hands on the wheel, if there is an emergency, is to keep them here all the time.

How to Turn Quickly and Safely. A quick turn can be made safely if it is done the right way. Here are some points that safe drivers use:

- Do not apply the brake while you are turning; it is very easy to lock your wheels while turning, which may cause you to skid out of control.
- Do not turn any more than needed to clear whatever is in your way; the more sharply you turn, the greater the chances of a skid or rollover.
- Be prepared to “counter-steer,” that is, to turn the wheel back in the other direction once you have passed whatever was in your path; think of emergency steering and counter-steering as two parts of one driving action.

Where to Steer. If an oncoming driver has drifted into your lane, a move to your right is best; if that driver realizes what has happened, the natural response will be to return to his or her own lane. If something is blocking your path, the best direction to steer will depend on the situation. If you have been using your mirrors, you will know which lane is empty and can be used safely. If the shoulder is clear, going right may be best, since no one is likely to be driving on the shoulder but someone may be passing you on the left. If you are blocked on both sides, moving to the right may still be best so that you do not force anyone into an opposing traffic lane and a possible head-on collision.

Leaving the Road. In some emergencies, you may have to drive off the road, which may be less risky than facing a collision with another vehicle. Most shoulders are strong enough to support the weight of a large vehicle and therefore offer an available escape route. If you do leave the road:

- Avoid braking until your speed has dropped to about 20 mph, then brake very gently to avoid skidding on a loose surface.

- Keep one set of wheels on the pavement, if possible, to help maintain control.
- Stay on the shoulder until your vehicle has come to a stop, then signal and check your mirrors before pulling back onto the road.

Returning to the Road. If you must return to the road before you can stop, hold the wheel tightly and turn sharply enough to get right back on the road safely; do not try to edge gradually back on because your tires might grab unexpectedly and you could lose control. When both front tires are on the paved surface, counter-steer immediately so the two turns are made as a single “steer–counter-steer” move.

2.17.2 – How to Stop Quickly and Safely

If somebody suddenly pulls out in front of you, your natural response is to hit the brakes, which is a good response if there is enough distance to stop and you use the brakes correctly. You should brake in a way that will keep your vehicle in a straight line and allow you to turn if it becomes necessary, using either the “controlled braking” method or the “stab braking” method.

Controlled Braking. Apply the brakes as hard as you can without locking the wheels and keep steering-wheel movements very small; if you need to make a larger steering adjustment or if the wheels lock, release the brakes and re-apply them as soon as you can.

Stab Braking. Apply your brakes fully and release them when the wheels lock up; as soon as the wheels start rolling, apply the brakes fully again, recognizing that it can take up to one second for the wheels to start rolling after you release the brakes, and that if you re-apply the brakes before they roll, the vehicle will not straighten out.

Do Not Jam on the Brakes. Emergency braking does not mean pushing down on the brake pedal as hard as you can, which will only keep the wheels locked and cause a skid; if the wheels are skidding, you cannot control the vehicle.

2.17.3 – Brake Failure

Brakes kept in good condition rarely fail, and most hydraulic brake failures occur for one of two reasons: loss of hydraulic pressure or brake fade on long hills (air brakes are discussed in Section 5).

Loss of Hydraulic Pressure. When the system will not build up pressure, the brake pedal will feel spongy or go to the floor; you should downshift to a lower gear to help slow the vehicle, pump the brake pedal to try to generate enough hydraulic pressure to stop, and use the parking or emergency brake—which is separate from the hydraulic system—while pressing or pulling the release so you can adjust pressure and keep the wheels from locking. While slowing, look for an escape route such as an open field, side street, or escape ramp, and turning uphill is a good way to slow and stop the vehicle; after stopping, ensure it does not roll backward by putting it in low gear, applying the parking brake, and if necessary rolling back into an obstacle that will stop it.

Brake Failure on Downgrades. Going slowly enough and braking properly almost always prevents brake failure on long downgrades, but once the brakes have failed you must look outside your vehicle for something to stop it. Your best option is an escape ramp, which will be marked by signs and is usually located a few miles from the top of the downgrade; some ramps use soft gravel to resist motion and stop the vehicle, while others turn uphill and use both the hill and soft gravel to hold it in place. Any driver who loses brakes going downhill should use an escape ramp if one is available, because not using it greatly increases

the chance of a serious crash; if no ramp is available, choose the least hazardous escape route you can, such as an open field or a side road that flattens out or turns uphill, and make the move as soon as you know your brakes do not work because waiting will let the vehicle gain speed and make it harder to stop.

2.17.4 – Tire Failure

Recognize Tire Failure. Quickly knowing you have a tire failure gives you more time to react, and having even a few extra seconds to remember what to do can help you. The major signs of tire failure are sound (the loud “bang” of a blowout, which you should assume is your tire), vibration (heavy thumping or vibration that may mean a flat, especially on a rear tire), and feel (heavy steering that may indicate a failed front tire, or sliding or “fishtailing” from a rear tire failure, which dual rear tires often prevent).

Respond to Tire Failure. When a tire fails, your vehicle is in danger and you must immediately hold the steering wheel firmly so a failed front tire does not twist it out of your hands, stay off the brake because braking with a failed tire can cause loss of control and you should wait until the vehicle has slowed before braking gently and pulling off the road, and after you stop, get out and check all the tires even if the vehicle seems to handle normally because with dual tires the only way to know one has failed may be by looking at it.